National Qualifications 2014

2014 Mathematics Paper 2

National 5

Finalised Marking Instructions

© Scottish Qualifications Authority 2014

The information in this publication may be reproduced to support SQA qualifications only on a non-commercial basis. If it is to be used for any other purposes written permission must be obtained from SQA's NQ Assessment team.

Where the publication includes materials from sources other than SQA (secondary copyright), this material should only be reproduced for the purposes of examination or assessment. If it needs to be reproduced for any other purpose it is the centre's responsibility to obtain the necessary copyright clearance. SQA's NQ Assessment team may be able to direct you to the secondary sources.

These Marking Instructions have been prepared by Examination Teams for use by SQA Appointed Markers when marking External Course Assessments. This publication must not be reproduced for commercial or trade purposes.

General Marking Principles for National 5 Mathematics

This information is provided to help you understand the general principles you must apply when marking candidate responses to questions in this Paper. These principles must be read in conjunction with the detailed marking instructions, which identify the key features required in candidate responses.

(a) Marks for each candidate response must always be assigned in line with these General Marking Principles and the Detailed Marking Instructions for this assessment.

(b) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.

(c) Credit must be assigned in accordance with the specific assessment guidelines.

(d) Candidates may use any mathematically correct method to answer questions except in cases where a particular method is specified or excluded.

(e) Working subsequent to an error must be followed through, with possible credit for the subsequent working, provided that the level of difficulty involved is approximately similar. Where, subsequent to an error, the working is easier, candidates lose the opportunity to gain credit.

(f) Where transcription errors occur, candidates would normally lose the opportunity to gain a processing mark.

(g) Scored out working which **has not been replaced** should be marked where still legible. However, if the scored out working **has been replaced**, only the work which has not been scored out should be marked.

(h) Where a candidate has made multiple attempts, mark all attempts and award the lowest mark.

(i) Unless specifically mentioned in the specific assessment guidelines, do not penalise:

- Working subsequent to a correct answer
- Correct working in the wrong part of a question
- Legitimate variations in solutions
- Bad form
- Repeated error within a question

Detailed Marking Instructions for each question

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
1.	**Ans: 590** •[1] know how to decrease by 15% •[2] know how to calculate roll •[3] carry out calculations correctly within a valid strategy and round to the nearest ten	**3**	 •[1] $\times 0{\cdot}85$ •[2] $964 \times 0{\cdot}85^3$ •[3] 590

Notes:

1. For an answer of 590 without working — award 3/3 — ✓✓✓

2. For an answer of 592 or 592·0165 without working — award 2/3 — ✓✓×

3. Where an incorrect percentage has been used, the working must be followed through to give the possibility of awarding 2/3 — ×✓✓

4. For an answer of 2460 $(964 \times 0{\cdot}85 \times 3)$ with working, — award 1/3 — ✓××

5. For an answer of 530 $(964 - 964 \times 0{\cdot}15 \times 3)$ with working, — award 1/3 — ✓××

6. For an answer of 430 $(964 \times 0{\cdot}15 \times 3)$ — award 0/3 — ×××

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
2.	**Ans: B (8, 4, 10), C (4, 0, 10)** •[1] state coordinates of B •[2] state coordinates of C	**2**	 •[1] (8, 4, 10) •[2] (4, 0, 10)

Notes:

1. For eg B(8, 4, 9) leading to C(4, 0, 9) — award 1/2 ×✓
2. The maximum mark available is 1/2 where
   (a) brackets are omitted
   (b) answers are given in component form

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
3.	(a)		**Ans: $5a + 3c = 158{\cdot}25$** • [1] construct equation	**1**	 • [1] $5a + 3c = 158{\cdot}25$
Notes: **1.** Accept variables other than a and c.					
	(b)		**Ans: $3a + 2c = 98$** • [1] construct equation	**1**	 • [1] $3a + 2c = 98$
Notes:					
	(c)		**Ans: Adult ticket costs £22·50** **Child ticket costs £15·25** • [1] evidence of scaling • [2] follow a valid strategy through to produce values for a and c • [3] calculate correct values for a and c • [4] communicate answers in money	**4**	 • [1] eg $\begin{matrix}10a+6c=316\cdot50\\9a+6c=294\end{matrix}$ • [2] values for a and c • [3] a = 22·5 and c = 15·25 • [4] Adult £22·50 Child £15·25
Notes: **1.** The fourth mark may only be awarded when **all** of the following are given in the final answer: the words "adult" and "child", the £ signs and **both** amounts written with two decimal figures.					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
4.	(a)	(i)	**Ans:** $\bar{x} = 56{\cdot}5$ • [1] calculate mean	**1**	• [1] $\bar{x} = 56{\cdot}5$
Notes: **1.** Do not accept 56·5 rounded to 57.					
		(ii)	**Ans:** s= **2·4** • [1] calculate $\left(x-\bar{x}\right)^2$ • [2] substitute into formula • [3] calculate standard deviation	**3**	• [1] 0·25, 0·25, 2·25, 2·25, 12·25, 12·25 • [2] $\sqrt{\frac{29{\cdot}5}{5}}$ • [3] 2·4(2….)
Notes: **1.** For use of alternative formula, award marks as follows: • [1] calculate $\sum x$ and $\sum x^2$ — • [1] 339, 19183 • [2] substitute into formula — • [2] $\sqrt{\frac{19183-\frac{339^2}{6}}{5}}$ • [3] calculate standard deviation — • [3] 2·4(2….) **2.** For correct answer without working — award 0/3					
	(b)		**Ans: No, standard deviation is greater** **OR No, times are more spread out** • [1] no, with valid explanation	**1**	• [1] e.g. No, standard deviation is greater
Notes: **1.** Answer must be consistent with answer to part (a)(ii). **2.** Accept "No, as 3·2 > 2·4" **3.** Only award the mark if it is clear that the reason is based on standard deviation **only**. **4.** Do not accept "No, times are less consistent" without further explanation.					

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
5.	**Ans: 3072 cm^3**	**3**	
	•^{1} state linear scale factor		•^{1} $\frac{24}{15}$ or 1·6
	•^{2} state volume scale factor		•^{2} $\left(\frac{24}{15}\right)^3$ or $1{\cdot}6^3$ (= 4·096)
	•^{3} calculate volume (calculation must involve a power of the scale factor) **and state correct units**		•^{3} 3072cm^3

Notes:

1. Some common answers

(a) 3072 award 2/3 ✓✓×

(b)1920cm^3 ($\left(\frac{24}{15}\right)^2 \times 750$) award 2/3 ✓×✓

(c)1200cm^3 ($\left(\frac{24}{15}\right) \times 750$) award 1/3 ✓××

(d)675000000cm^3 ($\left(\frac{24}{15}\right) \times 750^3$) award 1/3 ✓××

(e)183cm^3 ($\left(\frac{15}{24}\right)^3 \times 750$) award 2/3 ×✓✓

(f) 933cm^3 ($\left(\frac{15}{24}\right)^3 \times 750 + 750$) award 2/3 ×✓✓

2. The third mark is not available where premature rounding leads to an incorrect answer.
eg 4·1 × 750 = 3075cm^3 award 2/3 ✓✓×

3. Alternative Method

•^{1} know how to find radius of smaller cylinder •^{1} $\sqrt{\frac{750}{15\pi}}$

•^{2} know how to find radius of larger cylinder •^{2} $\left(\frac{24}{15}\right) \times \sqrt{\frac{750}{15\pi}}$

•^{3} calculate volume **and state correct units** •^{3} 3072cm^3

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
6.	**Ans: no, with valid reason.**	**4**	
	• [1] valid strategy		• [1] use Converse of Pythagoras' Theorem eg 110^2 and 85^2+75^2
	• [2] evaluation		• [2] 12 100 and 12 850
	• [3] comparison		• [3] e.g. $110^2 \neq 75^2+85^2$
	• [4] valid conclusion		• [4] No, since not right angled

Notes:

1. For alternative methods, award marks as follows:

(i)
- • [1] valid strategy — • [1] use Pythagoras' Theorem eg 85^2+75^2
- • [2] evaluation — • [2] h = 113·36
- • [3] comparison — • [3] e.g. $113\cdot36 > 110$
- • [4] valid conclusion — • [4] No, since not right angled

(ii)
- • [1] valid strategy — • [1] substitute correctly into cosine rule eg $\frac{85^2+75^2-110^2}{2\times85\times75}$
- • [2] evaluation — • [2] $86\cdot6°$
- • [3] comparison — • [3] $86\cdot6° < 90°$
- • [4] valid conclusion — • [4] No, since not right angled

2. There must be an explicit comparison for the award of the third mark.

eg $\sqrt{85^2+75^2} = 113\cdot36.$

No, since not right angled. award 3/4 ✓✓×✓

3. Conclusion **must** involve reference to "not a right angle".

eg $110^2 = 85^2+75^2 \rightarrow 12100 \neq 12850.$

No, Hightown is not due north of Lowtown. award 3/4 ✓✓✓×

4. The final mark is not available where the candidate's only conclusion is an invalid statement involving the word bearing.

eg "No, Hightown is on a bearing of 87° from Lowtown, not 90°"

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
7.			**Ans: 150 cm^3**	**5**	
			• [1] substitute correctly into formula for volume of cone		• [1] $\frac{1}{3}\times\pi\times4^2\times15$ $(=251\cdot32....)$
			• [2] substitute correctly into formula for volume of sphere **or** hemisphere		• [2] $\frac{4}{3}\times\pi\times3\cdot7^3$ $(=212\cdot17....)$ or $\frac{1}{2}\times\frac{4}{3}\times\pi\times3\cdot7^3$ $(=106\cdot08....)$
			• [3] know to subtract volume of **hemisphere** from volume of cone		• [3] evidence
			• [4] carry out all calculations correctly (must involve difference or sum of two volume calculations)		• [4] 145·24....
			• [5] round **final answer** to 2 significant figures		• [5] 150 (cm^3)

Notes:

1. Accept variations in π.

2. Some common answers (working must be shown):

(i) $39\ (\frac{1}{3}\times\pi\times4^2\times15-\frac{4}{3}\times\pi\times3\cdot7^3)$ award 4/5 ✓✓×✓✓

(ii) $120\ (\frac{1}{3}\times\pi\times4^2\times15-\frac{1}{2}\times\frac{4}{3}\times\pi\times4^3)$ award 4/5 ✓×✓✓✓

(iii) $110\ (\frac{1}{3}\times\pi\times3\cdot7^2\times15-\frac{1}{2}\times\frac{4}{3}\times\pi\times3\cdot7^3)$ award 4/5 ×✓✓✓✓

(iv) $160\ (\frac{1}{3}\times\pi\times8^2\times15-\frac{1}{2}\times\frac{4}{3}\times\pi\times7\cdot4^3)$ award 4/5 ×✓✓✓✓

(v) $360\ (\frac{1}{3}\times\pi\times4^2\times15+\frac{1}{2}\times\frac{4}{3}\times\pi\times3\cdot7^3)$ award 4/5 ✓✓×✓✓

(vi) $460\ (\frac{1}{3}\times\pi\times4^2\times15+\frac{4}{3}\times\pi\times3\cdot7^3)$ award 4/5 ✓✓×✓✓

(vii) $80\ (\frac{1}{3}\times\pi\times3\cdot7^2\times15-\frac{1}{2}\times\frac{4}{3}\times\pi\times4^3)$ award 3/5 ××✓✓✓

(viii) $250\ (\frac{1}{3}\times\pi\times4^2\times15)$ award 2/5 ✓×××✓

3. The final mark is only available where answers to all intermediate steps involve at least three significant figures.

eg $251\cdot32-106\cdot08=250-110=140$ award 4/5 ✓✓✓✓×

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
8.	**Ans:** $\mathbf{5n^4}$ • [1] simplify powers in numerator • [2] cancel constants • [3] eliminate n from denominator	**3**	• [1] $10n^6$ • [2] $\frac{5n^6}{n^2}$ • [3] $5n^4$

Notes:

1. For $5n^4$ without working — award 3/3
2. For a final answer of $\frac{5n^4}{1}$ — award 2/3 ✓×✓
3. For an answer of $5n^3$
   - (a) (i) $\frac{10n^5}{2n^2}=5n^3$ — award 2/3 ×✓✓
   - (ii) $\frac{10n^6}{2n^2}=5n^3$ — award 2/3 ✓✓×
   - (b) (i) $\frac{n^4\times10}{2n}=\frac{n^4\times10}{2}=5n^3$ — award 1/3 ✓×✓
   - (ii) $5n^3$ without working — award 1/3

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
9.	**Ans:** $\mathbf{\frac{4x-15}{x(x+5)}}$ • [1] correct common denominator (or correct numerator) • [2] consistent numerator (or denominator) • [3] simplify	**3**	• [1] $x(x+5)$ or $7x-3(x+5)$ • [2] $\frac{7x-3(x+5)}{x(x+5)}$ • [3] $\frac{4x-15}{x(x+5)}$

Notes:

1. Correct answer without working — award 3/3

2. For $\frac{7x}{x(x+5)}-\frac{3(x+5)}{x(x+5)}$ — award 2/3 ✓✓×

3. For subsequent incorrect working, the final mark is not available.

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
10.	(a)		**Ans: 84·8°** •1 substitute correctly into cosine rule •2 calculate cos B correctly •3 calculate angle ABC correctly	**3**	•1 $\cos B = \frac{8^2 + 11^2 - 13^2}{2 \times 8 \times 11}$ •2 cos B = 0·09……. •3 85 or 84·8…….
Notes: **1.** For 1·48 (uses RAD) or 94·2 (uses GRAD), with working award 3/3 **2.** The 2nd mark can be awarded for $\cos^{-1}\left(\frac{16}{176}\right)$					
	(b)		**Ans: 155·2°** •1 know how to calculate the angle •2 correctly calculate the angle within a valid strategy	**2**	•1 360 – 120 – [answer to (a)] or equivalent •2 155·2
Notes:					

Question			Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
11.			**Ans:** $a = \frac{2(s-ut)}{t^2}$ •1 subtract ut •2 multiply by 2 •3 divide by t^2	**3**	•1 $s - ut = \frac{1}{2}at^2$ •2 $2(s-ut) = at^2$ •3 $a = \frac{2(s-ut)}{t^2}$
Notes: **1.** Correct answer without working award 3/3 **2.** For subsequent incorrect working, the final mark is not available. **3.** For $a = \frac{s-ut}{\frac{1}{2}t^2}$ award 2/3					

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
12.	**Ans: $x°$ = 63°, 297°** •[1] rearrange equation •[2] find one value of x •[3] find another value of x	**3**	•[1] $\cos x = \frac{5}{11}$ •[2] x = 63 •[3] x = 297

Notes:

1. The 2nd angle must be consistent with the first angle.

2. For x = 1·1,358·9 (uses RAD), award 3/3 (with working), award 2/3 (without working)

3. For x = 70, 290 (uses GRAD), award 3/3 (with working),award 2/3 (without working)

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •
13.	**Ans: 151·3 m^2** •[1] know how to find area of segment •[2] know to express sector as a fraction of a circle •[3] know how to find area of sector •[4] know how to calculate area of triangle •[5] carry out all calculations correctly within a valid strategy	**5**	•[1] evidence of e.g. major sector + triangle or circle − minor sector + triangle •[2] $\frac{310}{360}$ **or** $\frac{50}{360}$ •[3] $\frac{310}{360}\times\pi\times7^2$ (= 132·56) **or** $\frac{50}{360}\times\pi\times7^2$ (= 21·38) •[4] $\frac{1}{2}\times7\times7\times\sin50$ (=18·77) •[5] 151·3 m^2

Notes for question 13 are on next page.

Question	Expected Answer(s) Give one mark for each •	Max Mark	Illustrations of evidence for awarding a mark at each •

Notes:

1. Accept variations in π; disregard premature or incorrect rounding of $\frac{310}{360}$ or $\frac{50}{360}$.

2. Use of RAD or GRAD (working must be shown)
(a) For 149·9 [uses GRAD] award 5/5
(b) Where the use of RAD leads to an answer of 126·1(−6·43+132·56) or 139·0(6·43+132·56) award 4/5

3. Some common answers (working must be shown):

56·6 $\left(\frac{310}{360}\times\pi\times14+\frac{1}{2}\times7\times7\times\sin50^\circ\right)$ award 4/5 ✓✓×✓✓

40·1 $\left(\frac{50}{360}\times\pi\times7^2+\frac{1}{2}\times7\times7\times\sin50^\circ\right)$ award 4/5 ×✓✓✓✓

2·6 $\left(\frac{50}{360}\times\pi\times7^2-\frac{1}{2}\times7\times7\times\sin50^\circ\right)$ award 4/5 ×✓✓✓✓

24·9 $\left(\frac{50}{360}\times\pi\times14+\frac{1}{2}\times7\times7\times\sin50^\circ\right)$ award 3/5 ×✓×✓✓

132·6 $\left(\frac{310}{360}\times\pi\times7^2\right)$ award 2/5 ×✓✓××

21·4 $\left(\frac{50}{360}\times\pi\times7^2\right)$ award 2/5 ×✓✓××

18·8 $\left(\frac{1}{2}\times7\times7\times\sin50^\circ\right)$ award 1/5 ××××✓×

153·9 $\left(\pi\times7^2\right)$ award 0/5

4. The fifth mark is only available when the area of triangle MON is calculated using trigonometry.

[END OF MARKING INSTRUCTIONS]